			Deterministic	LLM-based										
Benchmark	Set	Initial F_{max} (MHz)	**(a) Fitted heuristic**	**(b) Sequential reasoning**			**(c) Branching exploration**			**(d) Knowledge base-guided exploration**			Functional pass (%)	Best ΔF_{max} MHz (%)
				Sonnet	Gemini	Luna	Sonnet	Gemini	Luna	Sonnet	Gemini	Luna		
boom_soc	K	48.2	91.0	27.0	68.7	29.3	89.0	20.2	85.9	89.9	91.6	102.2	100	+49.3 (102.2)
vexriscv_re-place	K	310.2	51.4	31.5	31.5	31.5	32.6	32.6	41.9	42.8	32.6	42.8	100	+159.5 (51.4)
amd_mini_isp	K	307.1	32.7	33.2	22.2	23.6	23.6	29.4	32.6	30.0	33.2	31.6	100	+102.0 (33.2)
logicnets_jscl	K	403.6	23.8	10.3	2.4	2.4	4.3	17.7	23.8	23.8	24.1	3.8	60	+97.2 (24.1)
finn_radioml	K	284.9	21.1	3.2	3.2	3.2	10.2	20.1	20.1	17.3	20.1	21.2	87	+60.5 (21.2)
rosetta_digit-recognition	K	367.0	15.9	0.0	0.0	0.0	2.1	14.3	15.0	10.7	11.7	15.7	100	+58.2 (15.9)
rosetta_optical-flow	K	324.9	4.8	0.1	0.1	0.1	5.6	5.6	5.6	3.9	4.9	4.8	100	+18.3 (5.6)
corescore_500_mod	K	344.2	0.0	9.3	0.7	0.7	7.4	9.3	9.3	9.3	9.3	9.3	100	+32.1 (9.3)
boom_soc_v2	R	77.2	14.8	9.4	9.9	9.5	26.4	16.5	16.6	9.0	22.6	25.8	100	+20.4 (26.4)
amd_mini-isp_v2	R	288.0	41.5	40.9	43.3	40.7	33.2	38.3	39.7	39.7	40.3	40.2	100	+124.7 (43.3)
rosetta_3d-rendering_v2	R	236.7	21.1	3.4	3.4	3.7	3.7	26.2	26.3	16.4	20.1	26.3	100	+62.2 (26.3)
vexriscv_re-place_v2	R	397.5	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	100	+1.7 (0.4)
vtr_mcml_v2	R	69.3	0.0	3.4	3.4	3.4	3.5	3.4	3.4	3.5	3.4	3.4	100	+2.4 (3.5)
fir_symmetric_systolic	O	405.7	0.0	11.4	11.4	11.4	10.1	10.0	11.4	10.1	10.6	10.1	100	+46.2 (11.4)
fir_systolic_transposed	O	355.5	0.0	0.1	0.1	0.1	1.6	1.6	2.3	1.6	3.3	2.6	100	+11.9 (3.3)
picoblazearray	O	350.8	0.0	1.5	1.9	2.0	1.5	1.5	1.5	1.3	1.5	1.5	100	+7.0 (2.0)
		Functional pass (%)	100.0	100.0	100.0	95.8	100.0	100.0	93.8	100.0	100.0	93.8		

F_{max} **improvement (%)**

0 10 20 30 40 50

Set key

K known (development set)
R held out, same family
O held out, out of family